SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	Chill Chill
Project / Product Name	KerjayaKu: a living career map for the AI era
Final Live Deployment URL	https://kerjayaku-demo.onrender.com/
Codebase Repository URL	https://github.com/Aloysius8895/xmumHack
Name (Leader)	Edixon Teo Zan Wei
Email & phone (Leader)	edixontzw@gmail.com | 010 880 9196

2. Project Description

[Maximum 250 words]

KerjayaKu is a 3D career-planning game that helps Malaysian students and fresh graduates answer one question: what should I learn next, and why? A learner describes their skills or uploads a CV, and DeepSeek Flash turns it into an editable skill profile with a reason for every skill. Learners can also paste a real job posting and plan against that exact role.

KerjayaKu then compares the profile with live job postings from LinkedIn and JSearch, JobStreet samples weighted to DOSM's occupation structure, TalentCorp's Critical Occupations List, and the Anthropic Economic Index, which shows where AI is already entering each task.

A rules engine, not the AI, ranks 22 careers with an explainable score (50% skill fit, 50% career outlook), shows the exact skill gap and the three skills that would raise the score most, and re-ranks everything under moderate and extreme AI-adoption scenarios, always labelled as estimates. The gaps become a "no-regret" roadmap: skills that matter in both futures come first, and every task has a time estimate, a free learning link from a checked list of 132 links, and the SDG target it advances.

Learners record progress on the web or through Telegram, and a short quiz confirms the learning before the roadmap and their career pet advance. The interface runs in English, Bahasa Melayu and Chinese. KerjayaKu supports human decisions: it never makes hiring decisions, and the AI cannot invent jobs, courses or statistics.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

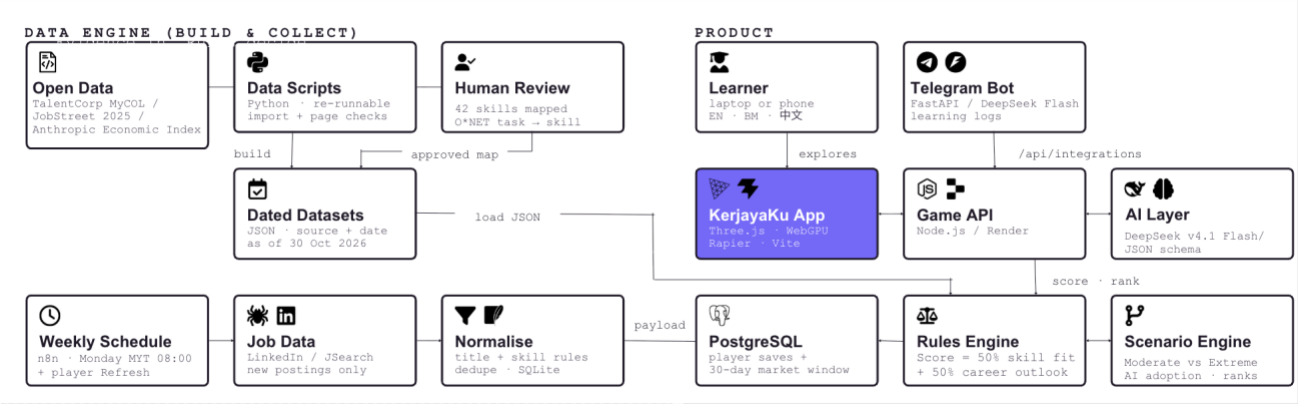


Figure 1. *KerjayaKu architecture: a data engine that collects and dates evidence, and a product in which code decides and AI only explains.*

One rule across the system. Evidence comes in with its source and date, a deterministic rules engine computes every score, rank and progress step, and the generative AI only reads unstructured input and writes explanations in a fixed JSON format.

Data engine (reference data). Re-runnable Python scripts import TalentCorp's Critical Occupations List (7 editions, 2015/16–2024/25) and AI Impact Study with page-level quotes, DOSM's quarterly occupation shares, 56,307 JobStreet 2024 and 9,554 JobStreet 2025 postings (aggregates only), and two releases of the Anthropic Economic Index mapped from O*NET tasks to our 67 catalogue skills. Each dataset is saved as dated JSON with a source manifest (URL, retrieval date, HTTP status, SHA-256).

Data engine (live job data). An n8n workflow triggers an incremental collection every Monday at 08:00 MYT, and a learner can press Refresh (30-minute cooldown). The collector reads LinkedIn public search and the JSearch API, normalises titles and skills with the same rules as the game, removes duplicates, fetches only new or changed postings, and places each posting in a MASCO stratum so samples are weighted to DOSM's occupation mix with 95% margins. Measured on the cloud profile, all 22 careers take 53 requests and 145 seconds. A failed run never replaces the last good snapshot. Months that our own collection has not reached yet are filled by a documented generator, marked mock in the data and in the careers panel, and replaced as real months accumulate.

Rules and scenario engine (Node.js). Matching, the explainable score, "how to raise it", the moderate and extreme scenarios, the No-Regret roadmap order, SDG mapping and capped progress are pure functions in src/game/server (scenarios.js, no-regret.js, sdg.js, game.js). They read only skill levels, never age, gender or other personal attributes, and are covered by 402 Node test cases.

Generative AI integration. The Game API calls DeepSeek Flash through its Anthropic-compatible endpoint with a forced JSON-schema tool call for six tasks: skill extraction from text or CV, job-posting analysis, roadmap wording, learning-log classification, quiz generation and mentor chat. Skill IDs and SDG targets are restricted to our own enumerations; learning links are chosen by index from the checked list, so the model cannot output a URL; malformed output gets one corrected attempt and is then rejected. API keys stay on the server, and an AI_MOCK demo mode returns canned answers if the provider is unavailable.

Database. PostgreSQL stores versioned player saves (game_saves) and shared documents such as the 30-day market window (app_documents), with optimistic version checks and a migration script; local JSON files are the offline fallback. The collector and the Telegram bot keep their own SQLite logs.

Frontend and channels. The 3D world uses Three.js (WebGPU with WebGL 2 fallback), Rapier physics and Vite, with career panels in vanilla JavaScript in three languages. A FastAPI Telegram bot posts learning logs to token-protected /api/integrations endpoints, and the same two-question quiz check runs on the web and in Telegram. The product runs as a Render web service built from the GitHub repository.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

Phase 1 delivered one working feature: a profile-based match. In the final round we rebuilt KerjayaKu around the six required capabilities and live evidence, in 24 commits from 30 October to 1 November (113 commits by 6 contributors since 20 October).

Built and integrated: incremental job-data collection with a 30-day window, Refresh and n8n scheduling; DOSM weighting with a national estimate and 95% margins; eight new careers selected from the data (22 in total, 67 skills); the scenario engine with an explainable score, "how to raise it" and a moderate/extreme switch; a No-Regret roadmap with weekly check-ins; 132 free learning links checked as reachable; and quiz-based learning verification on the web and in Telegram.

Problems we solved and pivots:

1. Hard-coded trends. We removed every hand-written heat and trend value and rebuilt the market panel on collected postings.

2. Misleading data. JobStreet's 2024→2025 change ranked chefs above software engineers because the two samples came from different collectors, so we kept it for display only and excluded it from the model.

3. Blocked sources. Indeed, Ricebowl and JobStreet return bot challenges; we chose not to bypass them and rely on LinkedIn public search and the JSearch API.

4. Identical scenarios. Our first outlook formula capped at 1, so both scenarios gave the same score; we corrected it and rebalanced the score to 50% fit and 50% outlook.

5. Box-ticking. Learners could tick tasks without learning, so progress now advances only after a short quiz.

6. Hosting change. When sponsor credits were withdrawn, we moved deployment to Render, which runs our Node.js server as a single long-running web service with no code changes.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	Your Response
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	**Phase 1 KPI:** median minutes to produce a valid three-step learning plan (a target role, at least two skill gaps and three actionable steps) from the same synthetic profile and job advert. Phase 1 target: 20% less time than manual planning, with at least 8 of 10 volunteers completing the journey unaided. **Phase 2 result (stated honestly):** the 10-person timed pilot has not been completed yet, so we do not claim a measured reduction. Our team walk-through estimate is 15–20 minutes manually (longer for an unfamiliar industry) against about 10 minutes in KerjayaKu, an expected 33–50% reduction against the 20% target, to be confirmed by the pilot. **Verified in the built prototype:** for the synthetic sample profile the engine returns every element of a valid plan in one journey: a ranked target (Data Analyst, 54/100 under moderate adoption), the three largest gaps with their point value (SQL +8.0, Statistics +7.2, Data cleaning +4.0) and an ordered roadmap in which each task carries a time estimate and a free learning link. 22 careers are ranked under two scenarios; 10 of them change rank when adoption is extreme, while the plan's core stays the same. All 67 skills have at least one free resource (132 links). **SDG contribution:** every career links to a UN target with a stated reason and every skill to a target; the learner's path reports progress towards SDG 4.4 (relevant skills for employment) and SDG 8.5 (productive employment) as intermediate outcomes, not job placement.

Metric Category	Your Response
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	**Reproducible engine log:** the numbers above come from running src/game/server/scenarios.js on the synthetic profile (code and output in Section 4.1). **Automated tests:** 402 Node test cases and 147 Pytest cases in tests/, including scenarios, No-Regret order, SDG mapping, real-data-only rules and learning-log validation. **Data provenance:** source manifests with URL, date, HTTP status and SHA-256; MyCOL quotes re-read from the PDF pages by import_official_signals.py; learning links checked by check-learning-resources.mjs (31 October 2026). **Human review of the weighting:** 200 postings checked by a team member, 82.5% placed in the right DOSM occupation group (rules 87%, AI 75%); the rules then adopted the reviewer's conventions. **Performance counters:** collection run for 22 careers measured at 53 requests and 145 seconds. **Development evidence:** Git history of 113 commits since 20 October 2026. **KPI pilot protocol:** same synthetic profile and advert, manual and KerjayaKu order counterbalanced, fixed validity rubric, reporting sample size and failures.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	In Malaysia 56% of professionals already use AI at work but only 26% have formal AI training (Hays Asia Salary Guide 2025), and MOSTI expects more than 600,000 workers to need reskilling within three to five years. KerjayaKu turns that pressure into one concrete next step per learner, free of charge. **Partners and reach:** university and TVET career centres can use it to advise whole cohorts with sourced, dated evidence instead of static lists; HR and learning-and-development teams can see which skills to train next; the skill gaps map to free resources and, next, to HRD Corp-claimable programmes. Three languages and a phone-friendly low-quality mode widen access. **Trust:** scores use skill levels only, scenarios are labelled as estimates, and the AI never decides for the learner.

Metric Category	Your Response
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	**Stack:** Render web service (3D app and API), PostgreSQL, DeepSeek Flash, n8n and a FastAPI Telegram bot, all configured through environment variables and documented in the README. **Unit cost (estimate):** AI about US$0.02 per active user per month (DeepSeek V4-Flash list price, US$0.14 / 0.28 per million input / output tokens, assuming 80k + 25k tokens per user); learning content US$0; hosting US$0 on Render's free instance during the pilot and about US$7 per month for an always-on Render Starter instance at launch, so the cost per user falls as usage grows. **Integration model:** a white-label web app for career centres, with token-protected integration endpoints (/api/integrations) so an institution's portal or chatbot can push learning records and read recommendations. **Privacy:** the prototype processes synthetic profiles only; before launch we will add explicit consent for CV analysis, remove names and contact details before text reaches the AI, host data in the region and support deletion requests under Malaysia's amended PDPA.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Scoring logic (src/game/server/scenarios.js, as deployed). Each skill's future value V combines hiring demand with AI exposure; the career score is computed in code and every term is shown to the learner.

```
adoption     D = 1 - e^(-k*H/tau)          H=3 years, tau=5
             moderate k=0.6 -> D=0.30      extreme k=3 -> D=0.83
skill value  V = T * (1 - a*D) * (1 + 0.5*u*D)
             T = hiring demand, a / u = automation / augmentation (Economic Index)
fit          = sum(w * min(cur,req)/req * V) / sum(w*V)     w = level x weight
outlook      = clamp(mean(V) * careerDemand / 2, 0, 1)
score        = round(100 * (0.5*fit + 0.5*outlook))
raise(skill) = 100 * 0.5 * w*V*(req-cur)/req / sum(w*V)     top 3 = "how to raise it"


export const WEIGHTS = { fit: 0.5, outlook: 0.5 }
const value = demand * (1 - automationLoss) * (1 + augmentationGain)
score: Math.round(100 * (WEIGHTS.fit * fit + WEIGHTS.outlook * outlook))
```

Run log on the synthetic profile (Excel 2, SQL 1, Python 1, data cleaning 2, data visualisation 1, communication 2; levels 0–3), 22 careers:

```
== moderate  k=0.6  adoption 0.302
#1 data-analyst    score 54  fit 0.564  outlook 0.521
   raise: sql +8.0  statistics +7.2  data-cleaning +4.0
#2 ml-engineer     score 54  fit 0.196  outlook 0.878
#3 ui-ux-designer  score 45  fit 0.111  outlook 0.799
#4 data-scientist  score 41  fit 0.268  outlook 0.558
== extreme   k=3    adoption 0.835
#1 data-analyst    score 52  fit 0.556  outlook 0.480
   raise: sql +8.1  statistics +7.6  data-cleaning +4.0
#2 ml-engineer     score 51  fit 0.191  outlook 0.835
rank changes between scenarios: 10 of 22 careers
   e.g. lawyer 14 -> 11, backend-developer 13 -> 19
```

What the proof shows. The score is fully decomposable (54 = 0.5 × 56.4% + 0.5 × 52.1%), the recommended next step is computed rather than generated (learning SQL to the required level is worth +8 points), and changing only the stated AI-adoption assumption moves 10 of 22 careers while the learner's top target and its three core gaps stay the same. That stable core is what the No-Regret roadmap teaches first, so the plan holds in both futures.

Learning verification. A learning record moves the roadmap only after the learner answers both questions of a two-question quiz generated about the logged topic (three options each); the correct answers stay on the server (src/game/server/log-validation.js, tests/game/server/log-validation.test.mjs).

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

1. Validate with partners (months 0–3). Run the timed pilot with an XMUM or Rotary-network partner career centre, publish the KPI result, and use joint research with XMUM to test whether the scores agree with career advisers' judgement.

2. Licensed data at volume (months 3–9). Replace public-page sampling with licensed job-data feeds and official partners (JSearch paid tiers, job boards, MYFutureJobs), and keep three or more comparable months so skill trends and confidence move from "current demand" to measured momentum. Corporate partners in the Rotary network can validate skill lists for their industries.

3. Institution-ready platform (months 6–12). Add accounts and per-institution dashboards, including a cohort skill-gap heatmap for HR and learning-and-development teams; serve the 3D assets from a CDN; move collection to a queue so dozens of careers and sources run in parallel; and map gaps to HRD Corp-claimable programmes.

4. Responsible scale. Consent, data minimisation, regional hosting and deletion under the amended PDPA, regular fairness checks on profile extraction, and a choice of AI provider for partners with data-residency needs.

Funding model. Free for students; institutions and employers pay a per-seat licence that covers data and hosting, while the AI cost stays near two US cents per active user per month.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable. All team-owned code, data work and 3D characters were created during the hackathon period (repository history from 20 October 2026). The 3D game foundation is third-party open-source work (Bruno Simon, folio-2025, MIT) and is declared in Section 7.2, not claimed as our own.

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Bruno Simon folio-2025 (3D world foundation)	Code / 3D assets	MIT; licence notice kept in the repository
Three.js, Rapier, Vite, Howler, camera-controls, Tweakpane, glTF-Transform	Code	MIT / Apache-2.0
GSAP	Code	GSAP Standard “no charge” licence
Anthropic TypeScript SDK (client for DeepSeek’s Anthropic-compatible API), pg, sharp, pdf-parse	Code	MIT / Apache-2.0
FastAPI, Uvicorn, HTTPX, python-dotenv, pytest	Code	MIT / BSD-3-Clause

DeepSeek Flash via DeepSeek API (live AI for game and Telegram bot)	Model / service	DeepSeek API terms; no training or fine-tuning by the team
AI coding assistants used during development: Replit Agent, OpenAI Codex, Anthropic Claude; ChatGPT for an AI-assisted review of the AI-exposure mapping	Model / tool	Provider terms; all output reviewed and committed by team members
Anthropic Economic Index (incl. O*NET task statements)	Dataset	CC BY (data); O*NET CC BY 4.0
TalentCorp Critical Occupations List 2015/16–2024/25 and AI Impact Study (Phase 2 Vol. 2)	Dataset / publication	Public publications, quoted with page references
DOSM job-market and occupation statistics; Techmap monthly totals	Dataset	Public releases, cited with source manifests
JobStreet Malaysia 2024 (Hugging Face azrai99/job-dataset)	Dataset	CC BY 4.0; aggregates only
JobStreet Malaysia 2025 (Hugging Face seemarn/jobstreet)	Dataset	No licence stated; aggregates only, no posting text redistributed
OpenJobData archived postings (Hugging Face scholarweave/Jobs-Dataset-v2)	Dataset	Dataset terms; FDE aggregates only
LinkedIn public job search; JSearch API (RapidAPI)	Data source / service	Public pages read without login, aggregates only; JSearch provider terms
WEF Future of Jobs Report 2025; UN SDG indicator framework	Publication	Cited, not redistributed
Free learning links (e.g. GeeksforGeeks, W3Schools, Khan Academy, OpenStax, MDN)	Linked resources	Linked only, not copied
Render, PostgreSQL, n8n, Telegram Bot API, GitHub	Services / tools	Provider terms; n8n Sustainable Use Licence

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable.

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

Signature: _____edixon_____ (Edixon Teo Zan Wei, Team Leader)

Date: 1 November 2026